MTH209: Endsem Exam - Question 2

1. Follow the instructions **exactly**.
2. If your code is not properly commented or horizontal/vertical spacing is missing, points will be deducted.
3. You are NOT allowed to use any external libraries in this question.

Question

Consider a random sample $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n \overset{iid}{\sim} N_2(\mu, \Sigma)$ where the $\mu \in \mathbb{R}^2$ and the covariance matrix is of the form

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma^2 \end{pmatrix} ,$$

Here, σ_i^2 is the marginal variance of the ith component and ρ is the correlation coefficient between the two components. Suppose each $\mathbf{X}_i = (X_{i1}, X_{i2})^\top$. A popular estimator of ρ is the sample correlation coefficient:

$$\hat{\rho} = \frac{\sum_{i=1}^n (X_{i1} - \bar{X}_1)(X_{i2} - \bar{X}_2)}{\sqrt{\sum_{i=1}^n (X_{i1} - \bar{X}_1)^2}\sqrt{\sum_{i=1}^n (X_{i2} - \bar{X}_2)^2}} , \quad \text{where} \quad \bar{X}_1 = \frac{1}{n}\sum_{i=1}^n X_{i1}, \qquad \bar{X}_2 = \frac{1}{n}\sum_{i=1}^n X_{i2}.$$

It is well known that

$$\sqrt{n}\left(\hat{\rho} - \rho\right) \overset{d}{\to} N\left(0, (1-\rho^2)^2\right) .$$

Consider testing $H_0 : \rho = 0$ against the alternative that $H_a : \rho \neq 0$. The goal of this question is to estimate the Type-I error of a statistical test for this hypothesis. However, this will be done many parts. For each part you have to submit a function following the appropriate instructions.

Functions to submit

1. Create a function `sim_normal` that takes μ and σ_1, σ_2 and ρ as input along with the number of samples and returns those many number of samples from $N(\mu, \Sigma)$:

```
# n: a positive integer denoting the same size
# mu: a 2-dimensional vector of means
# sigma1: first marginal standard deviation
# sigma2: second marginal standard deviation
# rho: correlation
sim_normal <- function(n, mu, sigma1, sigma2, rho)
{
  samples <- matrix(0, nrow = n, ncol = 2)
  ...
  ...
  # returned samples must be an nx2 matrix
  # each row is a draw from normal
```

```
  return(samples)
}
```

2. Create a function `test_stat` that given inputs, returns the test statistic for this test. For the variance normalization, you may use $\hat{\rho}$ as a plug-in for ρ. **HINT:** The function `cor` on an $n \times 2$ matrix returns $\hat{\rho}$.

```
# X: an nx2 dimensional matrix from Normal distribution
test_stat <- function(X)
{
  ...
  ...
  statistic <- ...
  return(statistic) # a single number
}
```

3. Create a function `type1_error` that takes in various inputs and returns the estimate of the Type I error of this test from repeated simulations.

```
# reps:  number of replications
# n: a positive integer denoting the same size
# mu: a 2-dimensional vector of means
# sigma1: first marginal standard deviation
# sigma2: second marginal standard deviation
# alpha: level of significance
type1_error <- function(reps, n, mu, sigma1, sigma2, alpha)
{
  track <- numeric(length = reps)
  ...
  ...
  ...
  error <- ...
  return(error) # one number
}
```

Submission Instructions

INSTRUCTIONS: copy and paste ALL three functions into a file called `question2.R` and upload it to mooKIT. Upload **ONLY the function** and nothing else. This question has 40 marks.